

Spatio-Temporal Hybridization, Open-World Forensic Signature Encoding, and Blockchain-Integrated Chain of Custody in Digital Media Forensics: A Unified Deep Learning Perspective

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Abstract—The rising prevalence of photorealistic synthetic media, referred to as deepfakes, has resulted in a worldwide decline in the trustworthiness of digital content. While the current state-of-the-art digital media forensics techniques have shown promising results for precise detection, they have shown significant limitations for open-world scenarios, where the manipulation process is unknown. This paper proposes a novel integrated framework for digital media forensics, aiming to resolve the following two problems: (i) precise detection of deepfakes using spatio-temporal deep learning, and (ii) open-world digital media characterization using compact signature encoding. This proposed framework includes a detection module, which employs a ResNeXt-LSTM hybrid deep neural network for the concurrent detection of spatial and temporal inconsistencies of the manipulated videos. Moreover, the proposed framework includes an open-world characterization module, which uses the EvidenX framework to encode the complete digital processing history of a digital media object as a 25-dimensional signature using denoising autoencoders and a quadruplet loss function. This proposed framework is secured using a blockchain-based chain of custody mechanism, which provides legal admissibility, security, and auditability. This proposed framework has shown promising results for the detection of manipulated videos using standard benchmarks, achieving accuracies greater than 95% and retrieval times of less than 400 ms for a database of one million media objects. This proposed framework suggests a paradigm shift in the field of digital media forensics, focusing on a more holistic approach, rather than a narrowly defined binary classification problem, by incorporating the concepts of explainable deep learning, classical signal processing, and blockchain-based data integrity verification.

Keywords: deepfake detection, media forensics, spatio-temporal modeling, open-world forensics, media signature encoding, blockchain, chain of custody, EvidenX, ResNeXt-LSTM.

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I. THE SOCIO-TECHNICAL LANDSCAPE OF DEEPFAKES AND MEDIA INTEGRITY

The recent rise of highly realistic fake media, referred to as "deepfakes," has disrupted the field of digital media forensics and information security. Traditionally, people have been relying on the authenticity of visual evidence based on the information embedded in the media files and conventional signal processing techniques. However, with the advent of new media synthesis techniques such as GANs and VAEs, the reliability of such techniques is diminishing. [1], [2]. Deepfakes refer to the unauthorized alteration of the facial look or voice of a person in videos or audio recordings. Deepfakes have mostly been used to spread disinformation, assist with the impersonation of individuals, or fuel political instability. [3].

The risks to society from deepfakes are multifaceted and range from the individual level to the media and democracy. The democratization of powerful media generation tools and the availability of high-performance computing resources have enabled the distribution of a significant amount of manipulated media on social media platforms, which is too large to be manually validated. [4], [5]. The sheer volume of online content—wherein there are about 14.1 billion images shared online on a daily basis, as well as more than 2 million hours of video content uploaded to YouTube on a minute-to-minute basis—has transformed a traditionally trustworthy human visual system into a potential liability that an adversary might exploit through AI-generated fabrication. [6], [7].

Digital evidence occupies a central position in contemporary jurisprudence, and preserving its integrity, provenance, and legal admissibility is of paramount importance [8]. This duty calls for technological measures that ensure transparency, explainability, and a strong chain of custody—a continuous audit trail of all individuals who have handled the evidence before it is presented to the court. Unlike physical artefacts, digital evidence is naturally malleable, reproducible, and alterable without detection, which further emphasizes the need for immutable audit trails that are not accessible to unauthorized parties.

The ability to produce synthetic media content using variational autoencoders (VAE) and generative adversarial networks (GANs) has facilitated the development of false identities and the embedding of genuine individuals in compromising or false settings, such as non-consensual intimate images, which marks one of the first recorded cases of deepfake technology in social media. [1]. Technologies like facial reenactment and expression transfer enable an adversary to manipulate the facial movements of promi-

nent public figures, thus creating the illusion that they are making statements that they did not actually make, an ability that has significant consequences for political stability. [9]. This ecosystem of synthetic disinformation has been exacerbated by the growth of large open-source data sets and the rapid progress of deep generative models, whose results are increasingly being repurposed through social media algorithmic pipelines in ways that actively work against detection. [10].

Among the most formidable challenges confronting digital forensics practitioners today is the so-called “open world” problem [11]. In contrast to the controlled environment of laboratory testing, in real-world operational forensic work, the histories of media files are often complex and partially unknown. The files could have been manipulated by unknown editing software, could have been subject to multiple generations of compression, and could have been shared across multiple social media platforms before being seized and analyzed by investigators. Forensic systems are typically designed primarily as binary classifiers, and are often tailored to work well for specific, known chains of manipulation, making them particularly prone to failure in the face of unfamiliar manipulation chains or post-processing techniques. [12], [13]. This paper addresses both the detection challenge and the open-world characterization problem as two inseparable dimensions of a unified forensic task.

II. TAXONOMY OF MANIPULATION TECHNIQUES AND FORENSIC PRINCIPLES

In digital media forensics, three major categories of video manipulation in deep fake media have been identified, each with its own challenge to video detection systems. In *Head puppetry*, the entire head and upper torso of the subject are completely synthesized to mimic the movements of another actor in the video. [3]. *Face swapping* involves the replacement of a subject’s facial region with a rendered donor face while preserving the underlying expressive motion [3]. *Lip synchronization* confines manipulation to the oral region, fabricating the appearance that a subject is articulating content they never produced [3].

At the image level, forensic analysis must additionally contend with inpainting, copy-move forgery, and composite splicing operations. AI-driven inpainting systems—including STTN [14], OPN [15], and GM-CNN [16]. They have the capability to restore the deleted or occluded areas of the images in the video frames, and they do so in a near-photorealistic manner. This, therefore, adds a level of complexity to the detection process. Copy-move attacks,

as the name suggests, use the internal regions of images to conceal or create evidence, whereas splicing involves the use of different materials.

Forensic detection is based on two underlying postulates. The first postulate states that any operation performed on an image file leaves behind latent artifacts, such as resampling irregularities and inconsistencies in local noise, that are embedded in the manipulated content. [3]. The second one recognizes the fact that current generative AI systems do not have an innate understanding of human biophysiology and physical limitations, which results in systematic and detectable violations such as abnormal blinking patterns. [17]. Unnatural head positions and inconsistencies in the biometric information within the ear region are also identified as signs of forgery [3]. Researchers are using these signal-level anomalies such as aliasing caused by spatial resampling and high-frequency ringing caused by upsampling as a dependable sign of forgery.

Moreover, empirical results have also revealed that traditional forensic practices, such as analyzing compression artifacts and examining image metadata, often failed to identify GAN-based forgeries that have been subjected to secondary processing operations such as spatial resampling and platform-specific re-encoding. This has, in turn, led to an overarching shift in the direction of developing more learning-based forensic practices that directly utilize raw data for deriving discriminatory characteristics. Even though initial deepfake technology showed signs of less attention being paid to biological plausibility, recent research has revealed that abnormal textural characteristics, bilateral asymmetry, and abnormal spectral energy characteristics in anatomical regions such as the forehead, perioral, and nasal bridge areas are significant forensic features. [18].

III. THE HYBRID RESNEXT-LSTM DETECTION ARCHITECTURE

The hybrid detection methodology described in the recent literature, with particular reference to the contribution by [19] [19], This work demonstrates the efficacy of a dual-pathway deep learning framework that combines spatial feature analysis and temporal sequence understanding. The proposed framework simultaneously discovers visual manipulation features and motion level anomalies by jointly using Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for appearance level and dynamic consistency understanding, respectively.

III-A. Spatial Feature Extraction via ResNeXt-50

The spatial analysis part of the detection model is achieved using the ResNeXt-50, which is a neural network structure that is highly acclaimed for its high representational ability. The ResNeXt network is an extension of the standard residual network, where grouped convolutional layers are used, referred to as *cardinality*—which enable the concurrent learning of a broader feature vocabulary without a commensurate increase in model parameters [19]. The architectural modularity of ResNeXt, as well as the availability of ImageNet pre-trained weight initializations, enables the use of transfer learning—a significant advantage since training data is often scarce.

The mathematical structure of the ResNeXt transformation block is expressed in terms of cardinality (C), denoting the number of parallel independent transformation pathways:

$$y = x + \sum_{i=1}^C \mathcal{T}_i(x) \quad (1)$$

where $\mathcal{T}_i$ Represents the i -th parallel transformation applied to the residual input x . During the process of fine-tuning, the network is conditioned to identify resolution-level inconsistencies that are embedded within the affine facial warp regions. These inconsistencies are systematically introduced as part of the face generation process [19]. The model attains higher sensitivity to textural discontinuities and boundary-transition anomalies by focusing on cropped facial subregions.

III-B. Temporal Modeling with Long Short-Term Memory

Studying individual video frames will not be enough to reliably discriminate deepfakes, particularly when considering high-quality synthesis where individual frames are indistinguishable from real content. Temporal anomalies, such as inter-frame jitter, unnatural motion trajectories, and discontinuities in appearance, are a class of forgery indicators that are exclusively detectable by analyzing the sequence. [20]. The use of a Long Short-Term Memory (LSTM) network is utilized to process the ordered sequence of spatial embedding features provided by the ResNeXt encoder, thus enabling the extraction of motion-level cues and the long-range dependencies that are usually difficult for generative models to reproduce with fidelity. [20], [19].

The Long Short-Term Memory (LSTM) module operates on the frame-level embedding sequences and detects the temporal discontinuities that occur as a result of the independent processing of the frames by the deepfake

generation system. This is equivalent to a detectable forensic signal. [19]. This capacity for temporal reasoning represents a decisive capability advantage over purely spatial detection architectures.

III-C. Preprocessing and Implementation Workflows

A well-designed preprocessing step is an important prerequisite for achieving an accurate model of detection. A series of steps are followed in the process, starting with the temporal segmentation of the input video into discrete frames [19], [2]. Next, face detection is performed on each of the frames, usually using the Viola-Jones detector or MTCNN, to isolate the region of interest, i.e., the faces, from the background. [19], [21]. This helps reduce computer-based complexity and difficulties. Subsequently, the faces are resized to match the spatial dimensions as required by the ResNeXt model.

Training set diversity and generalization capacity are expanded through a suite of data augmentation operations [19]. These changes include a random horizontal reflection, stochastic changes in brightness and contrast, synthetic motion blur, and the addition of JPEG degradation. Exposing it to these varying degradation mechanisms allows it to build robustness to changes in real-world image quality and capture conditions.

IV. EVIDENX: AN OPEN-WORLD FORENSIC INTELLIGENCE FRAMEWORK

While the ResNeXt-LSTM model shows promise for binary classification of deepfakes under conducive conditions, for effective forensic analysis, a far broader ability is required, one which can identify the complete digital provenance of a media object across its entire processing history, without any a priori knowledge of the tools used for its manipulation. This need for a broader ability has led to the development of a unified multimodal digital forensics framework known as *EvidenX* [11].

EvidenX This process goes beyond binary detection by offering a complete description of the digital lifecycle of any media artifact. Spatial, temporal, and structural features are combined in the form of a compact latent representation, termed the "media signature," embedded in a 25-dimensional signature space.[11]. Architecturally analogous to Siamese networks, *EvidenX* places media samples within a structured metric space such that objects with equivalent processing histories are proximal to one another, irrespective of whether the manipulation toolchain was seen during the training phase.

IV-A. Architecture Overview

The encoding process starts with the extraction of a heterogeneous feature set that includes both content-level semantic representations and format-level structural descriptors. These features are then mapped by the encoder, producing a compact signature embedding. $\mathcal{Z}$ that represents the full processing identity of the media object.

IV-B. Media Signature Encoder

The encoder $f_\theta : X \rightarrow \mathcal{Z}$ The model is designed to impose metrically consistent geometry on the signature manifold, where samples of the media data that share the same processing history y_1 are in closer proximity to each other than those that have different histories y_3 . This constraint is formally expressed as:

$$d(f_\theta(x_1), f_\theta(x_2)) < d(f_\theta(x_1), f_\theta(x_3)) \quad (2)$$

where $d(\cdot, \cdot)$ measures Euclidean separation within $\mathcal{Z}$. The encoder is trained using a Siamese-inspired architecture [22] governed by a quadruplet loss function [23]:

$$L_q = \max(d(z, z_s)^2 - d(z, z_{n1})^2 + m_1, 0) + \max(d(z, z_s)^2 - d(z_{n1}, z_{n2})^2 + m_2, 0) \quad (3)$$

where z and z_s denote anchor and co-class samples respectively, z_{n1} and z_{n2} denote samples drawn from two distinct negative classes, and m_1, m_2 are margin boundary hyperparameters. To equip the encoder with generalization capability over unobserved processing categories, a Denoising Autoencoder (DAE) branch [24] is incorporated, contributing an additional reconstruction objective computed uniformly across all quadruplet members:

$$L_r = \|x - \hat{x}\|_2^2 + \|x_s - \hat{x}_s\|_2^2 + \|x_{n1} - \hat{x}_{n1}\|_2^2 + \|x_{n2} - \hat{x}_{n2}\|_2^2 \quad (4)$$

The actual aggregate training objective imposes a simultaneous constraint on both discriminative metric alignment and generative reconstruction—this dual constraint is what equips the network with open-world generalization ability.

V. ENGINE MODALITIES AND SIGNAL PROCESSING METHODOLOGY

The *EvidenX* framework is organized as a collection of specialized media analysis engines, each of which is designed to identify a particular type of media manipulation evidence. Together, the analysis engines provide a comprehensive, multi-modal perspective on media evidence that is not possible with any one type of analysis.

V-A. Visual Forgery Detection via MesoNet

The MesoNet engine is designed to detect mesoscopic visual anomalies contained in a single frame of a video, using a MesoInception-4 architecture, where dilated convolutional layers are added to the Inception blocks.[21]. This configuration exhibits particular effectiveness in detecting Face2Face reenactment and deepfake-generated forgeries characterized by insufficient temporal coherence or aberrant micro-texture patterns [9]. By operating at a middle representational scale, intermediate between the level of pixel detail and the level of semantic meaning, MesoNet maintains its sensitivity to the evidence of forgery that would be overlooked by macro-scale feature extractors, and yet is more resistant to post-processing degradation than detectors using solely pixel-level analysis.

V-B. Compression Artifact Analysis via ELA

Error Level Analysis (ELA) is a signal processing-based method to identify regions within the image where re-compression has taken place at a quality level different from the rest of the image [25]. It does so by re-compressing the query image at a specified quality level, say 90%, and then determining the normalized discrepancy map between the original and re-compressed query image:

$$D = |I_{\text{orig}} - I_{\text{resave}}| \quad (5)$$

Pixels whose error magnitudes exceed the average for the entire image have been identified as possible areas for localized editing. This method can provide an alternative forensic clue for deep learning classifiers while ensuring that regional inconsistencies are based on sound signal processing principles.

V-C. Audio LSTM Engine

The audio forensic engine is designed to identify synthetic or manipulated speech by extracting Mel-Frequency Cepstral Coefficients (MFCCs) alongside their first- and second-order temporal derivatives [26], [27]. A Bidirectional LSTM processes the sequence of the temporal features to detect unusual patterns of prosody, such as unusual patterns of fundamental frequency, unusual phoneme transition dynamics, and spectral artifacts resulting from the use of vocoders, which are common in text-to-speech systems [26]. The combination of MFCC features with a BiLSTM architecture has demonstrated promising empirical performance in uncovering artifacts that arise in the neural vocoder stage, which is a common problem in modern GAN-based speech synthesis systems.

V-D. Container and Metadata Analysis

The video container analysis module parses file-level structural information via atomic tree representations of con-

tainer formats (e.g., the `ftyp` and `moov` atoms of the MP4 container specification), enabling reliable source software attribution at an accuracy of 97.6% [28]. This structural analysis is supplemented by JPEG header entropy counting and cross-field metadata consistency verification [29], which can expose discrepancies between a file’s declared origin and its actual processing lineage.

Another forensic aspect to consider is that of EXIF data, which involves analyzing device model IDs, GPS coordinates, and software source information. In cases where EXIF data conflicts with container metadata, it is likely that tampering occurred.

V-E. Network Forensics Branch

It has been integrated with a PCAP file analysis module that can process the data gathered from the network transmissions. The packets can be reconstructed in order to rebuild the entire network layer lifecycle of the evidentiary media [30]. This ability can help identify unauthorized data exfiltration events, anomalies at the protocol layer, and illicit access events that might be legally relevant for chain-of-custody documentation.

VI. EXPLAINABILITY AND FORENSIC VERIFICATION

Traditionally, deep learning models have been black-box classifiers that provide a definitive yes or no verdict on the authenticity of the image without offering any explanation for the decision-making process. This has resulted in the steady decline of the trust that legal practitioners have in AI-generated forensic results. [5]. The frameworks examined in this paper approach interpretability through two complementary pathways: spatial attention visualization and frequency-domain analytical verification.

VI-A. Attention-Based Learning and Grad-CAM

Attention-augmented convolutional architectures generate spatial weighting maps that selectively emphasize discriminative facial subregions rather than treating all image locations equivalently [31]. What Grad-CAM does is to represent the decision of the model as an image, so the important weights of the decision are represented visually. This provides the examiners with an understandable view of the decision. [5]. The activation concentrations are mostly grouped around the supraorbital ridge, the eyes, the nose, and the mouth, which are the most vulnerable areas to the visible artifacts of the generative models currently in use.

Combining the results of the domain experts’ reviews with the attention maps produced by the AI model has proven

to be especially valuable when dealing with borderline cases. Our empirical results suggest that 60% of the errors produced by the model occur when the confidence margin is below 0.5, which is exactly the interval where the additional value of the experts’ visual inspection is the greatest. [5].

VI-B. Frequency-Domain Mathematical Verification

Complementary forensic verification can be accomplished through the utilization of traditional signal processing techniques in the spectral domain. By employing the two-dimensional Discrete Fourier Transform (DFT), the unusual patterns of energy distribution in the frequency domain can be uncovered within the cropped area of the image, as these patterns have been associated with the employment of GANs in image synthesis. Additionally, the statistical behavior of the distribution of AC coefficients can be examined through the employment of Discrete Cosine Transform (DCT) β coefficients conforms to the expected Laplace model [18].

It has also been observed that frequency-domain-based methods are more likely to generalize better across various architectures compared to purely spatial-domain-based methods. This is because the artifacts introduced by GAN-based synthesis methods have a uniform pattern across various architectures, which is not observed in residuals [32]. This property renders spectral analysis particularly valuable for open-world forensic settings in which the underlying generator architecture remains unknown.

VII. EMPIRICAL PERFORMANCE AND BENCHMARK VALIDATION

VII-A. Deepfake Detection Benchmarks

The ResNeXt-LSTM hybrid architecture has undergone systematic validation across a range of established benchmark datasets, yielding strong and consistently reproducible detection performance.

Performance on the Celeb-DF dataset is particularly noteworthy. It is regarded as one of the most technically challenging benchmarks due to its complex synthesis process, which minimizes the visibility of splicing boundaries and color space differences between the manipulated regions and the background [1]. As the model has been able to achieve 91% accuracy using the Celeb-DF dataset, this demonstrates that the combination of cardinality-based spatial analysis and gated temporal analysis can reveal traces of manipulations that are imperceptible

However, the performance of various detection methods depends on the targeted forensic feature. Methods based

TABLE I
DEEPPFAKE DETECTION PERFORMANCE COMPARISON BY DATASET

Dataset	Acc.	Key Characteristics
DFDC	95.7%	Real-world diversity, noisy labels, variable quality
FaceForensics++	94.9%	Standardized, multiple methods [10]
DFD	91.0%	Google/Jigsaw curated dataset
Celeb-DF	91.0%	High-quality, minimal visible artifacts

TABLE II
PERFORMANCE COMPARISON OF INDIVIDUAL FORENSIC DETECTION STRATEGIES

Detector Type	Acc. (S1)	Primary Forensic Cue
Mouth-Based	94.21%	Lip/mouth movement inconsistencies
Blink [17]	82.15%	Eyelid opening/closing dynamics
Combined Fusion	97.00%	Spatio-temporal and physiological cues [34]

on the analysis of perioral motion achieve high accuracy because of the consistency of artifacts in the dynamics of the lip region. Lightweight models such as MesoNet [21], while computationally favorable, exhibit reduced efficacy against high-fidelity refined deepfakes. Capsule-based forensic networks [33] deliver improved modeling of spatial part-relationship structures but encounter scalability challenges at dataset scale.

VII-B. Individual Forensic Detector Comparison

These results demonstrate the main idea behind ensemble learning: if you combine different and complementary feature spaces, the accuracy will always be higher than that of the single-modality detector. The fusion score of 97% represents a significant improvement over the best single-modality detector.

VII-C. Open-World Retrieval: EVA-7K and In-painting Results

The generalization capability of the EvidenX signature encoder on an open world was evaluated on the EVA-7K benchmark set, which consists of 6,860 videos created with four tools, both commercial and free software, for video editing: Adobe Premiere, Avidemux, Kdenlive, and ffmpeg [28]. Leave-one-out cross-validation experiments confirm that signatures encoding media processed by held-out,

unknown toolchains—notably *Avidemux*—are correctly co-localized with signatures from known but related toolchains such as *ffmpeg*, owing to shared low-level encoding library foundations [11]. Furthermore, AI-driven manipulations produced by STTN [14], OPN [15], and GM-CNN [16] these are clearly distinguishable from the software-based edits made in the signature area, indicating that the encoder is successfully detecting the distinct statistical signatures of human-directed versus AI-directed processing.

VII-D. Social Media Sharing Chain Detection

The Media4Community experimental protocol employed the R-SMUD dataset (35,100 images) and the FODB dataset (23,106 images) [35], [36]. We collected the shared or re-propagated images from the social networks Facebook, Twitter, WhatsApp, Instagram, and Telegram. Among the interesting results, we highlight the temporal generalization capability of the proposed system. In fact, the signature encoder learned on the 2017 Facebook-distributed images was able to show high discriminative performance on the 2020 Facebook images in the FODB evaluation set [37]. This demonstrates that the signature representation learned by the model is actually picking up on the underlying structural and statistical fingerprint specific to the platform’s compression and processing setup, rather than the actual features of the content. In other words, it is capable of true temporal generalization.

VII-E. System Scalability

TABLE III
RETRIEVAL LATENCY USING MILVUS HNSW INDEX

Database Size	Operation	Latency (s)
6,480 items	Search	0.057
10 ⁶ items	Search	0.333
1 item	Feature Extraction	0.5–10.0

The retrieval of a single media signature from a database of one million items in less than 0.4 seconds illustrates the applicability of the *EvidenX* system to real-world digital infrastructures [11]. The fast lookup is made possible through the use of the Milvus vector similarity engine and its Hierarchical Navigable Small World (HNSW) indexing algorithm to provide the computational foundation for internet-scale forensic searches.

VIII. MULTI-MODULE FORENSIC PLATFORM: INTEGRATED CAPABILITIES

A comprehensive digital forensic investigation requires more than a purpose-built binary deepfake detector. The *EvidenX* platform aggregates various specialized forensic and utility tools in a single system to provide forensic examiners with a multidimensional perspective on media authenticity.

The system is powered by a FastAPI-based backend with a traditional JavaScript-based frontend to offer an interactive web-based analytics experience. The results of the analytics are collected into a digitally signed PDF report that is shared via a secure link, offering a clear chain of communication between the labs and the judiciary. The results of the detection are visualized as a heat map, with areas more likely to have been manipulated colored red and areas assumed to be authentic colored green.

IX. BLOCKCHAIN FOR DIGITAL EVIDENCE INTEGRITY AND CHAIN OF CUSTODY

As deepfake detection technology becomes more advanced as a forensic tool, there is a greater need to ensure a legal chain of custody for digital evidence. Digital evidence must be traced through a standard, immutable, and fully auditable path from its initial discovery to its final appearance in the courtroom. [8].

IX-A. Limitations of Conventional Evidence Management

The legacy forms of digital evidence management, such as access logs, paperless processes, and centralized databases, are, by their very nature, easy prey for insider attacks, human mistakes, and targeted cyber attacks. If the integrity of the evidence record is breached, the absence of a clear audit trail can result in critical, material information being omitted from the trial process. Tangibility, transferability, and the ability to modify digital information without any noticeable physical signs are the hallmarks of the importance of the aforementioned forms of vulnerability.

IX-B. Hybrid Blockchain Architecture: Forensic-Chain

However, the Forensic-Chain architecture has addressed these constraints through the implementation of the permissioned hybrid blockchain architecture using the Hyperledger Fabric and Composer frameworks [8]. The access to the evidentiary record is restricted to the specific group of participants such as law enforcement investigators, prosecutorial officers, magistrates, and certified forensic examiners. All the activities involving the creation, transfer, and access

of the evidence are recorded as immutable blocks in the blockchain technology system.

The system architecture is organized around four interdependent functional layers:

- **Participants:** Credentialed actors authorized to submit, transfer, or retrieve evidentiary records.
- **Fundamental Modules:** Programmatic interfaces through which participants query evidence metadata and verify provenance.
- **Smart Contracts:** Self-executing rule enforcement mechanisms that apply validation logic and consensus requirements, precluding unauthorized modification.
- **Distributed Evidence Store:** A partitioned, permission-gated storage layer secured by authorization tokens that enforce access control at the evidentiary unit level.

IX-C. Evidence Management Functions

The evidence lifecycle within Forensic-Chain is governed by three core operational functions [8]:

- 1) **Evidence Creation:** Upon submission of a unique evidence identifier and associated descriptive metadata, the system computes a SHA-256 digest of the digital artifact and instantiates a new, write-protected block—provided no prior record for that identifier already exists within the chain.
- 2) **Evidence Transfer:** Transfers custodial authority to a designated recipient address, contingent upon verified confirmation that the initiating party holds current legal possession of the item.
- 3) **Evidence Display:** Confirms the continued existence and cryptographic integrity of the evidence record prior to returning its associated information to requesting authorized parties.

By utilizing SHA-256 hashing, a permanent audit trail on an Ethereum-compatible network (such as Sepolia via Infura), the framework ensures mathematical integrity of evidence by catching any attempt to modify a block retroactively instantly [8]. The result is an entirely traceable evidence lifecycle, from initial collection to every touch point to court submission, meeting today's standards for admissibility.

IX-D. Integration with Forensic Detection Outputs

The Blockchain layer is not used alone. Instead, it is used side by side with the AI-based Forensic Analysis Pipeline. Once an EvidenX-based analysis is completed, a set of

TABLE IV
CORE FORENSIC MODULES OF THE EVIDENX PLATFORM

Module	Functionality
EXIF Metadata	Analyzes camera data, GPS coordinates, and software editing traces
Error Level Analysis	Uses JPEG re-compression heatmaps to highlight edited regions [25]
Copy-Move Detection	Identifies duplicated areas within a single image
Audio LSTM	MFCC-based BiLSTM analysis of speech prosody and GAN synthesis artifacts [26]
CNN Classification	Detects deepfakes and blending artifacts via MesoNet [21]
Signature Encoding	Open-world media fingerprinting via denoising autoencoder [11]
Blockchain Hashing	Secures evidence integrity using SHA-256 on-chain logging [8]

forensic report results is available, which includes ELA-based heatmap visualizations, CNN-based classification confidence levels, 25-dimensional media signatures, as well as EXIF data. The cryptographic hash of a combination of the report results and associated media artifact is then added to the Blockchain. If anyone tries to modify or alter the report or original media later on, the hashes will no longer match, which will then be detected by all the participants on the network. This is how a new standard is set for ensuring data integrity.

X. CHALLENGES AND OPEN RE-SEARCH DIRECTIONS

Despite the high detection accuracy provided by the architectures discussed in this paper, the field of deepfake forensics remains plagued by various challenges that the current systems have yet to effectively address.

X-A. Generalization Under Distribution Shift

The generative synthesis methods, including both GANs and diffusion models, are under constant evolution. This means that the detection methods are always under threat of new types of media emerging from architectures that are not included in the training data [5], [38]. Detection accuracy degrades markedly when models encounter adverse imaging conditions, including low spatial resolution, nonstandard illumination, and aggressive lossy compression [2]. Though many studies have pointed out the phenomenon of inter-type generalization, there is a significant drop in performance when using cross-dataset evaluation, which raises some basic questions regarding the domain transferability of current detection systems [18].

The *EvidenX* technique utilizes a processing fingerprint-based encoding technique. This technique shifts the focus of manipulation detection to a more general set of statistical

signatures, moving away from specific patterns associated with particular tools. This offers a degree of relief from the challenge [11]. Still, there are also flaws and limitations to the technique. As the variety of manipulation techniques increases, making them more similar to each other in terms of statistical signatures, the ability to differentiate between learned signatures in the latent space can continue to deteriorate.

X-B. Adversarial Attacks

Adversarial attacks—deliberate perturbations of deepfake content specifically engineered to circumvent targeted detectors—represent an escalating threat category [33], [32]. These attacks differ in the information available to the attacker, where white-box attacks assume that the attacker has access to the detector’s design, whereas black-box attacks are based purely on observable information or indirect statistical methods of inferring the detector’s operation. The development of ensemble-based detection systems showing robust adversarial robustness remains a research challenge.

X-C. Multimodal and Cross-Lingual Deepfakes

Contemporary deepfake operations are increasingly characterized by coordinated multi-modal fabrication, wherein manipulated video content is paired with synthetically generated audio and, in some instances, fabricated supporting text or metadata—constituting sophisticated disinformation operations that exploit the cross-modal nature of human perception [39], [38]. Detection systems designed to focus on a singular modality struggle to adapt to multi-front fabrication attempts. *EvidenX*’s BiLSTM-powered audio analysis module represents a critical first step in the direction of multi-modal forensic analysis support. However, the development of fully integrated multi-modal reasoning

models capable of assessing cross-modal coherence between audio, visual, and text streams is an open research problem.

X-D. Legal and Ethical Dimensions

But beyond all of that tech stuff, putting AI-powered forensic tools in a courtroom also raises a lot of big governance questions. Like, what happens if you get a false positive and how does that impact a suspect, or how do you ensure that you're using evidence that humans can understand and stand behind, and how do you avoid a situation where bad actors exploit openly shared detection methodologies? Tools like Grad-CAM and spectral analysis help make sense of all of that, but a robust governance model around AI-driven digital evidence in a courtroom is still in its infancy.

XI. FUTURE RESEARCH DIRECTIONS

Looking ahead, the most promising directions for advancing the field include the following areas.

Frequency-Aware and Diffusion-Era Detection. The advent of diffusion probabilistic models as a prominent model for high-fidelity image and video synthesis has created a new and ever-expanding family of forensic targets that have spectral properties that are inherently dissimilar from the spectral properties of GAN-synthesized media. The architectures are designed specifically to detect the spectral fingerprints that are distinctive of diffusion-based media generation [32]—which diverge substantially from those produced by adversarial training regimes—represent a critical near-term research priority.

Large Multimodal Models for Forensic Reasoning. The use of Large Multimodal Models (LMMs) as forensic reasoning engines with the capability to provide structured natural language explanations of the identified anomalies by the AI system is an area with considerable potential to improve the accessibility and judicial value of automated forensic analysis [40]. Forensic language model that is able to articulate, in terms that legal practitioners and jurors can understand, the explicit anatomical and computational rationale for a determination of deepfake would greatly enhance the prospects for ensuring that a determination of a legal issue by an AI model meets legal standards of admissibility.

Federated Forensic Learning. The issue of privacy and data sovereignty is a major impediment to the centralized aggregation of training data for forensic purposes. However, model generalization is contingent on access to a variety of data distributions. Federated learning frameworks offer a promising avenue to address these issues while training machine learning models without requiring access to sensitive evidence data.

Real-Time Verification at Platform Scale. However, if the ability for forensic verification were integrated directly into the content distribution and upload infrastructure, this would allow for preemptive authenticity assessment of the content before distribution to the public audiences. This would dramatically change the operational paradigm for the practice of digital media forensics from the traditional reactive approach to the more proactive approach of prevention. The sub-second retrieval latency that has been demonstrated for the Milvus-indexed database infrastructure suggests that this could be accomplished in real-time *EvidenX* system indicate that such deployment is technically feasible at the scale of major digital platforms [11].

XII. CONCLUSION

The current study proposes a comprehensive framework for the field of digital media forensics, divided into two associated paradigms: accurate deepfake detection through the application of spatio-temporal hybrid CNN-RNN architectures, and open-world media forensic provenance characterization through the derivation of compact learned media signatures. The proposed ResNeXt-LSTM-based detection model achieves detection accuracy above 90% for benchmark datasets, including Celeb-DF and DFDC [19], while the *EvidenX* signature encoding system supports toolchain-agnostic processing history characterization at sub-400 ms retrieval latency across one-million-item corpora [11].

Both of these investigative capabilities are integrated into a blockchain-secured chain-of-custody framework to ensure that all outputs of AI-generated forensics satisfy the criteria of legal admissibility, verifiability, and post-hoc resistance [8]. The inclusion of explainable artificial intelligence methodologies, which comprise Grad-CAM activation visualization techniques and frequency domain spectral verification, provides an added layer of confidence that the basis upon which a computer-based determination is made can be explained to legal practitioners or independently verified.

As the barriers to production of deep fakes continue to erode and the generative technology gets closer to, or possibly surpasses, the thresholds of current detection methodologies, continued investment in multi-modal frequency and adversarial robust forensic techniques will be critical to maintaining the integrity of the digital information landscape. Although the frameworks analyzed within this paper demonstrate a significant improvement in methodology, the adversarial nature of the forensic landscape dictates a continued push for innovation in

methodology as opposed to resting on current benchmark achievements.

Ultimately, the maintenance of authenticity and verifiability in digital media can be conceived of as a social and institutional imperative akin to the technological imperative. The technology created by the research community to aid in the investigation of crimes of cyber harassment must not only be effective in the lab, it must also foster a level of trust and understanding within the legal community, the journalistic community, and the citizenry it hopes to protect.

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